Provider & Practice Directory Update Pipeline

HealthLynked challenge — Option C: working prototype + production architecture.

Executive summary

Healthcare provider data decays constantly — providers move, practices merge or rebrand, phones and suites change, clinicians retire. Maintaining a directory by hand is expensive and doesn't scale. I built a **repeatable, cost-efficient pipeline** that keeps a provider **and** practice directory accurate, and I can **measure** its accuracy rather than assert it.

The thesis in one line: most provider and practice updates already sit in free, authoritative U.S. government datasets (NPPES + CMS), so the optimal pipeline is a **deterministic-first funnel** that resolves the bulk of records at ≈$0 marginal cost and reserves web scraping, LLMs, and human reviewers for the small residual where structured sources are silent or in conflict.

Three things set this submission apart:

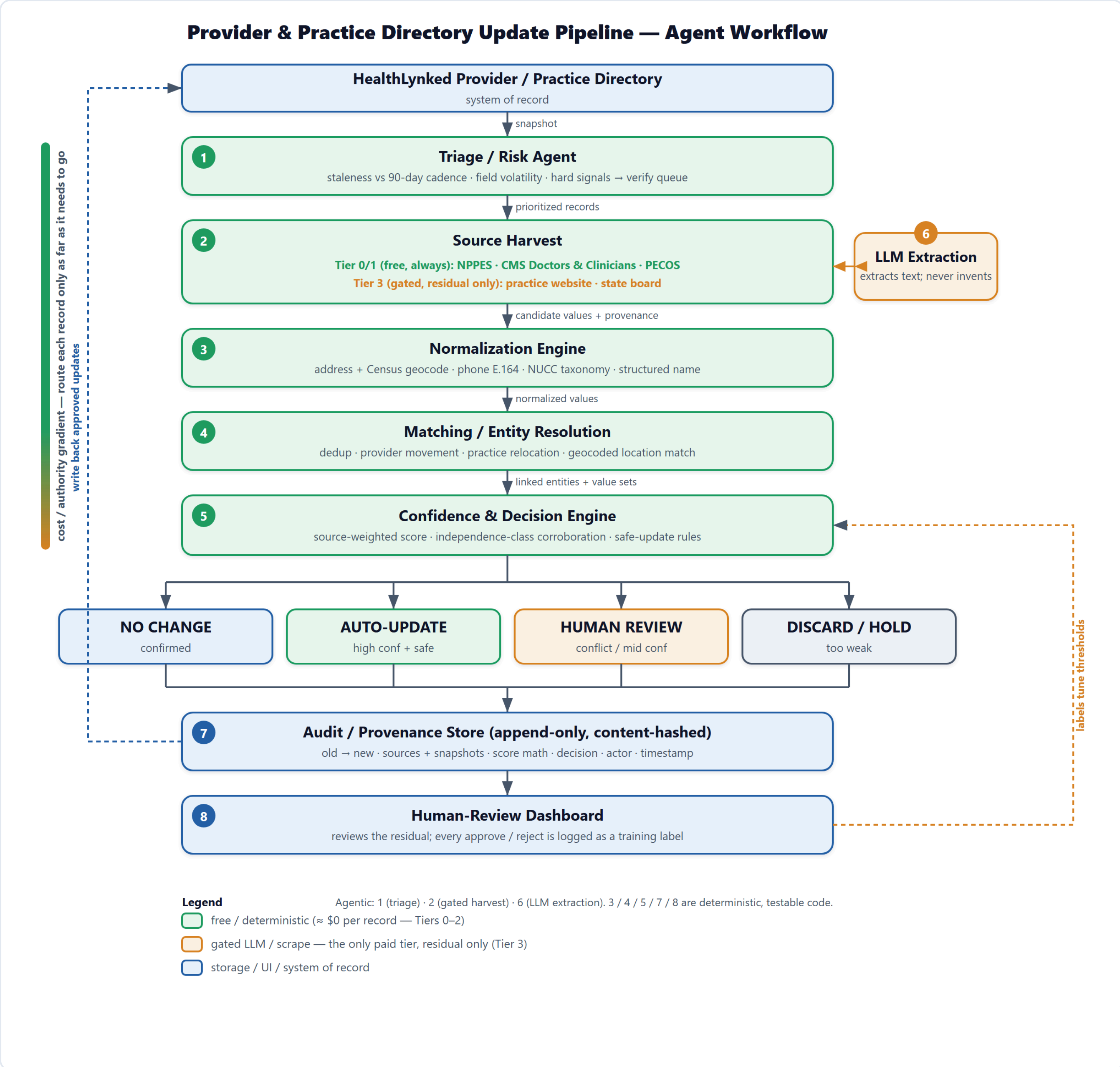
1. **It runs.** A one-command prototype reconciles the brief's exact `HL_001` example, emits the brief's exact JSON schema, and exercises every evaluation criterion end-to-end — offline, deterministically, with no API keys.
2. **Accuracy is measured.** Because the government sources are versioned monthly, I replay a past snapshot against a later one and report real precision, recall, and a calibration curve. A representative 5,000-record backtest shows **≈98% recall, ≈97% auto-update precision, 100% routing accuracy** (Brier ≈ 0.05).
3. **The timing is a legal mandate.** The **REAL Health Providers Act** (signed 2026-02-03) will require Medicare Advantage plans to verify every directory field every 90 days, remove departed providers within 5 days, and **publicly report an accuracy score** from PY2028. A pipeline that emits a defensible, auditable accuracy score is the compliance product HealthLynked's payer customers will be required to buy.

1. What the evaluation rewards, and the design choice that earns it

Criterion	Design choice
Accuracy	Anchor on authoritative structured sources (NPPES/CMS) before any LLM/scrape; never auto-update on a single weak source — **and prove it** with a historical-snapshot backtest.
Scalability	National bulk files, embarrassingly parallel by state/ZIP; blocking keeps matching sub-quadratic; Splink-ready for 100M+.
Cost efficiency	A free deterministic funnel resolves the large majority; LLM/scrape/human touch only the residual. Flat cost in volume.
Practicality	Free OSS + free government APIs; no exotic infra; a 2–3 person team can run it.
Explainability	Every recommendation carries the scoring math and the source snapshots that drove it.
Data quality	Canonical normalization for address (+ Census geocode), phone (E.164), specialty (NUCC taxonomy), and name.
Source reliability	Explicit per-source, per-field reliability weights; **independence-aware** corroboration; conflict baked into the score.
Human-review design	Risk-tiered routing so only true conflicts/low-confidence reach a human; a working review dashboard.
Audit trail	Append-only, content-hashed event store; every change links to source evidence and the confidence computation.

2. Architecture

A deterministic-first funnel routes each record only as far down a **cost/authority gradient** as it needs to go. The expensive tiers (LLM extraction, human review) only ever see records the free tiers couldn't resolve — that single choice is what keeps the cost curve flat as volume grows.



Naming the boxes "agents" is fair, but most are plain deterministic code: only triage, the gated harvest, and LLM extraction are agentic; normalization, matching, scoring, and the audit store are ordinary, testable functions. That honesty is a feature — it is what makes the system cheap and reproducible.

3. Data sources & reliability (free first)

Source	Authoritative for	Access	Cost
NPPES NPI Registry	Identity, NPI, taxonomy→specialty, practice/mailing address, phone, deactivation. Type-1 = providers, **Type-2 = practices** (both halves of the brief).	Free API (no key) + monthly bulk	$0
CMS Doctors & Clinicians (`mj5m-pzi6`)	Group/practice affiliation (PAC ID), practice locations, active enrollment.	Free API + bulk	$0
Historical NPPES/CMS snapshots (NBER mirror)	Ground truth for the backtest — what really changed month to month.	Free	$0
US Census Geocoder	Address standardization/validation + coordinates.	Free API (no key)	$0
State medical boards	License status (active/inactive/disciplinary).	Per-state scrape	scrape only
Practice website	Current phone/address/roster — strong corroborator.	Scrape + LLM (Tier 3 only)	$ (residual only)

Paid vendors (Google Places, Melissa, commercial provider data) are deliberately **avoided** on the default path — they are the easy way to blow the cost budget the brief explicitly penalizes.

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4. Confidence scoring (the formula)

For a field `f` and candidate value `v` asserted by sources `S`:

```
field_conf(f, v) = R(v) · A(v)
  A(v) = Σ_{s:val=v} w(s,f)·ϕ(s)  /  Σ_{s∈S} w(s,f)·ϕ(s)      (agreement share)
  R(v) = 1 − Π_{s:val=v} (1 − w(s,f)·ϕ(s))                  (noisy-OR strength)
```

- `w(s,f)` is a **per-field reliability weight** for the source class; `ϕ(s)` is a **freshness** factor that decays stale source data.
- `A` drags confidence toward the middle when sources disagree (genuine conflict → ambiguous score → human review). `R` rewards multiple independent confirmations.
- **Independence is enforced.** NPPES and CMS are *not* independent — both are ultimately self-reported by the provider to CMS — so same-class sources are de-correlated (the 2nd+ source in a class contributes only `1−ρ` of its weight), and the corroboration gate requires agreement across **≥2 distinct independence classes**: provider-self-reported gov (NPPES), claims/enrollment gov (CMS), practice web, and regulatory board.

The brief's conflict example falls straight out: the practice website and CMS show a new address while NPPES still shows the old one, so the address confidence collapses to ~0.58 and the record is routed to a human — while the phone, agreed by two sources, scores 0.88.

5. Decision & safe-update rules

Field **risk classes** govern thresholds and corroboration:

Class	Fields	Rule
Low	phone, suite, website	auto-update at `conf ≥ 0.85`, ≥1 class
Medium	full address (move), specialty, practice name, affiliation	auto-update at `conf ≥ 0.90` **and ≥2 independent classes**
High	provider name, **NPI**, active status, practice merge/close	**never silent** — always routed to a human

```
if no field differs                           → NO_CHANGE
elif a high-risk source contradicts current → HUMAN_REVIEW   (dissent safety net)
elif any changed field is in conflict       → HUMAN_REVIEW
elif a high-risk field changed              → HUMAN_REVIEW
elif min(conf) ≥ threshold and corroborated → AUTO_UPDATE
elif min(conf) ≥ τ_low                      → HUMAN_REVIEW
else                                        → DISCARD / HOLD
```

Two safety choices worth calling out:

- **NPI never auto-updates.** An NPI that maps to a different entity is a data-entry error or a merged duplicate → a repair/dedup ticket, never a silent field write.
- **Downgrades are never silently ignored.** If a credible source reports a provider *inactive* while the record (backed by a stale stronger source) still shows *active*, the record is routed to review even though nothing "changed" — missing a deactivation is exactly the trust-eroding error the brief cares about.

6. Normalization, matching & lifecycle signals

- **Normalize:** address → canonical components + Census geocode (suite handled separately so a suite change never looks like a move); phone → E.164; specialty → NUCC taxonomy code; name → structured parts with credentials pulled out.
- **Match (Fellegi–Sunter, Splink-ready):** dedup by blocking on NPI and `(soundex(name), ZIP)`; **practice-location matching uses geocoded proximity**, not fragile string similarity.
- **Lifecycle signals:** the pipeline labels *provider movement*, *practice relocation*, *closure* (deactivation), *rebrand*, and *NPI repair* as structured events — not guesses.

7. Triage — the cost lever

The cheapest verification is the one you skip. Each record is risk-scored so a cycle actively harvests only the top of the queue; everything else rides the free monthly batch diff.

```
risk = w1·staleness(days vs 90-day cadence) + w2·field_volatility
     + w3·hard_signal(deactivation / diff / bounce) + w4·directory_importance
```

8. Validation & calibration — *measured, not asserted*

Because the authoritative sources are versioned monthly, I replay snapshot **T** as "the directory" and score the pipeline's proposals against snapshot **T+Δ** as ground truth — labels for free, at national scale. The same harness is a regression test and produces exactly the accuracy score the REAL Act now mandates.

The repo ships three fidelity levels behind one metric implementation: a fast offline stand-in (CI/regression), a **synthetic two-snapshot world at scale** (with a realistic fraction of stale/wrong sources), and a builder that diffs two real NBER NPPES snapshots. A representative `--scale 5000` run:

Metric	Result
Recall of real changes	**97.7%**
Auto-update precision (1 − wrong silent writes)	**96.8%**
Routing accuracy	**100%**
Brier score / ECE	**0.05 / 0.08** (well-calibrated)

Confidence is then calibrated (isotonic/Platt) so a published `0.90` means ~90% of such updates are correct — what makes the audit trail and the mandated accuracy score defensible to a payer's compliance auditor.

One honest limit: a snapshot backtest measures agreement with the *authoritative record*, and NPPES contact data is self-reported and can lag the real world. So for the fields whose ground truth lives off the government grid — a working phone, a current suite — production also holds out a small **hand-labeled gold set** (a few hundred records verified by phone/web) to measure on-the-ground accuracy and keep the Tier-3 enrichment honest. The backtest covers identity / taxonomy / affiliation / deactivation at national scale; the gold set covers contact accuracy in depth.

9. Cost model — per 1,000 records

Stage	Records / 1,000	Subtotal
Tier 0/1 batch reconcile (NPPES+CMS bulk, amortized)	1,000	~$0.50
NPPES API + Census geocode (targeted, free)	~300	$0
Tier 3 web fetch + LLM extraction (residual only)	~100	$1–3
Human review (true conflicts only)	~30–50	$30–50
Total		**≈ $32–54 / 1,000 ≈ $0.03–0.05 per record**

Cost is **dominated by human review** — which the entire architecture exists to minimize — and is **flat in volume** because Tiers 0–1 are free batch. A naive "LLM-every-record + paid address API" approach lands 10–20× higher and is less accurate.

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10. Scalability

The national NPPES file is ≈8–9M NPIs (~1 GB) and full reconciliation is embarrassingly parallel (partition by state/ZIP). The per-record pipeline is a pure map — drop it behind a worker pool or queue. Matching blocks so it never goes quadratic and swaps to Splink (100M+ on Spark). Live calls retry with backoff and cache on disk, so re-runs and the 90-day re-verify cadence don't re-hit the network.

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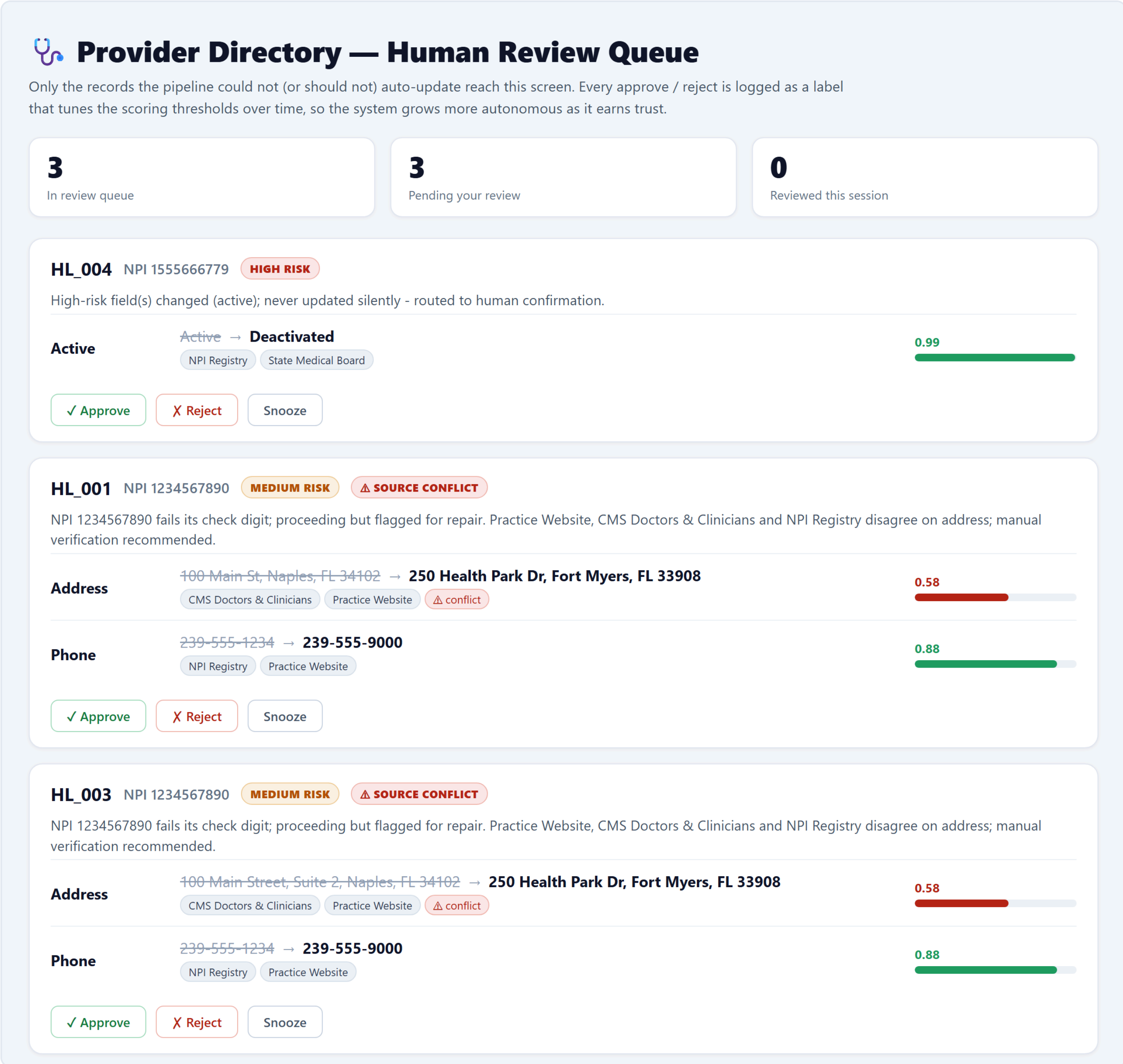
11. Human review & audit trail

A lean Streamlit dashboard shows one card per record needing a human — each proposed change side-by-side with its confidence, supporting sources, and conflict flag, with the queue sorted so the highest-risk and conflicting records surface first. Every approve/reject is logged as a **training label** that tunes thresholds over time, so the system grows more autonomous as it earns trust.



The review queue on the demo data: the high-risk deactivation (HL_004) is surfaced first, then the two address conflicts where the practice website and CMS disagree with NPPES — each field showing its calibrated confidence, supporting sources, and conflict flag. A reviewer approves, rejects, or snoozes per record; the green/amber/red confidence bars make the safe calls obvious at a glance.

Every proposed or applied change writes one immutable, **content-hashed** audit event: old→new value, the source snapshots (with weights and freshness), the score breakdown, the decision, the actor, and the pipeline version — so any update is reproducible and defensible months later.

12. Roadmap (also the 3-month consulting plan)

Phase	Weeks	Deliverable
0 — MVP (this submission)	—	Runnable funnel, schema-compliant output, audit log, dashboard, backtest.
1 — Structured backbone	1–4	Bulk NPPES+CMS ingestion; full-directory monthly reconcile; backtest harness on real snapshots from day one.
2 — Matching & triage	4–8	Splink dedup/movement in production; risk-based triage; weekly incrementals + deactivation diffs.
3 — Residual enrichment	8–12	Gated Tier-3 agent (web/board) with budget caps + caching; dashboard live; threshold tuning from reviewer labels.

KPIs, measured continuously by the backtest: auto-update precision ≥ 99% on the production threshold, human-review ≤ 5%, cost ≤ $0.05/record, freshness within the 90-day window, plus a published, calibrated accuracy score.

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13. Risks & mitigations

Risk	Mitigation
NPPES address is self-reported and can lag	Treat as strong-but-not-sole; require corroboration for moves; Tier-3 confirms.
State-board scraping is brittle (50 sites)	Build incrementally, highest-volume states first; board is a corroborator, not a hard dependency.
LLM hallucination	LLM only *extracts* from fetched text; the value must still clear the scoring gate; snapshot stored.
Over-aggressive auto-update	Conservative thresholds, min-field-confidence decision, high-risk always human-confirmed, full reversibility via the audit log.

Appendix — the prototype

```
pip install -e ".[dev,calibration]"
directory-pipeline demo                        # end-to-end offline showcase
directory-pipeline backtest --scale 5000       # the measured accuracy curve
streamlit run dashboard/app.py                 # human-review dashboard
```

The demo reconciles a 6-record sample (including the brief's `HL_001` and a Type-2 practice record), prints triage, recommendations, lifecycle signals, duplicate detection, and a measured backtest — all logged to a content-hashed audit store.

directory-pipeline - demo

```
======================================================================
 Provider & Practice Directory Update Pipeline - offline demo
======================================================================
 6 sample records | sources: NPPES + CMS + web/board (fixtures)

Verify queue (triage — most at-risk first):
    [0.63] HL_004  (1980d since last verify (> 90d window))
    [0.51] HL_002  (1490d since last verify (> 90d window))
    [0.48] HL_006  (1320d since last verify (> 90d window))
    [0.41] HL_001  (1016d since last verify (> 90d window))
    [0.41] HL_003  (1016d since last verify (> 90d window))
    [0.19] HL_005  (104d since last verify (> 90d window))

[?] HL_001  John Smith, MD
    action     : human_review  (overall 0.73)
    reason     : NPI 1234567890 fails its check digit; proceeding but flagged for repair. Practice Website, CMS Doctors & Clinicians and NPI…
    change     : address: '100 Main St, Naples, FL 34102' -> '250 Health Park Dr, Fort Myers, FL 33908'  (conf 0.58; Practice Website, CMS D…
    change     : phone: '239-555-1234' -> '239-555-9000'  (conf 0.88; NPI Registry, Practice Website)
    signal     : provider movement: 100 Main St, Naples, FL 34102 -> 250 Health Park Dr, Fort Myers, FL 33908
    geocode    : address validated (US Census)

[+] HL_002  Jane A Doe, NP
    action     : auto_update  (overall 0.96)
    reason     : NPI 1002003006 recovered by name/org search (agreement 1.00). Updated address, phone confirmed by independent reliable sour…
    change     : address: '300 1st St, Fort Myers, FL 33901' -> '501 HEALTHPARK BLVD, FORT MYERS, FL 33908'  (conf 0.96; Practice Website, N…
    change     : phone: '239-555-2000' -> '239-555-7777'  (conf 0.96; NPI Registry, Practice Website, CMS Doctors & Clinicians)
    signal     : provider movement: 300 1st St, Fort Myers, FL 33901 -> 501 HEALTHPARK BLVD, FORT MYERS, FL 33908
    geocode    : address validated (US Census)

[?] HL_003  John Smith M.D.
    action     : human_review  (overall 0.73)
    reason     : NPI 1234567890 fails its check digit; proceeding but flagged for repair. Practice Website, CMS Doctors & Clinicians and NPI…
    change     : address: '100 Main Street, Suite 2, Naples, FL 34102' -> '250 Health Park Dr, Fort Myers, FL 33908'  (conf 0.58; Practice W…
    change     : phone: '239-555-1234' -> '239-555-9000'  (conf 0.88; NPI Registry, Practice Website)
    signal     : provider movement: 100 Main Street, Suite 2, Naples, FL 34102 -> 250 Health Park Dr, Fort Myers, FL 33908
    geocode    : address validated (US Census)

[?] HL_004  Robert Lee, MD
    action     : human_review  (overall 0.99)
    reason     : High-risk field(s) changed (active); never updated silently - routed to human confirmation.
    change     : active: 'Active' -> 'Deactivated'  (conf 0.99; NPI Registry, State Medical Board)
    signal     : provider inactive/retired
    geocode    : address validated (US Census)

[=] HL_005  Maria Garcia, MD
    action     : no_change  (overall 0.89)
    reason     : All checked fields match authoritative sources; record confirmed accurate.
    geocode    : address validated (US Census)

[+] HL_006  Gulf Coast Women's Health
    action     : auto_update  (overall 0.94)
    reason     : Updated address, phone confirmed by independent reliable sources (min confidence 0.94 >= 0.90).
    change     : address: '850 6th Ave N, Naples, FL 34102' -> '3300 PINE RIDGE RD, NAPLES, FL 34109'  (conf 0.95; Practice Website, NPI Reg…
    change     : phone: '239-555-6000' -> '239-555-6100'  (conf 0.94; Practice Website, NPI Registry, CMS Doctors & Clinicians)
    signal     : practice relocation: 850 6th Ave N, Naples, FL 34102 -> 3300 PINE RIDGE RD, NAPLES, FL 34109
    geocode    : address validated (US Census)

Duplicate clusters detected:
    [DUP_001] HL_001, HL_003  (score 1.00; identical NPI 1234567890)

Audit log written to: audit_log.jsonl  (10 events)
Records awaiting human review: 3  (5 field-level events)

----------------------------------------------------------------------
Backtest report (historical-snapshot replay)
====================================================
  records evaluated        : 6
  true field changes       : 9
  detected (correct/all)   : 9/9

  recall                   : 100.0%
  detection precision      : 100.0%
  AUTO-UPDATE precision    : 100.0%  (4/4)
  routing accuracy         : 100.0%  (6/6)

  Brier score (lower=better): 0.0434
```

```
  expected calib. error     : 0.1422

  NOTE: only 9 scored changes — too few for a stable
        calibration curve. Run `directory-pipeline backtest --scale 5000`
        for the statistically meaningful curve (isotonic in production).
----------------------------------------------------------------------
Launch the human-review dashboard with:  streamlit run dashboard/app.py
```

Solo submission. All code is free/OSS; all data sources are free and authoritative.